

• Globally asynchronous locally synchronous (GALS) implementation

Its advantages are as follows:

- Smaller die area
- Deeply timing closure
- Faster frequency and latency
- Easy configuration
- Shorter schedules
- Automated verification
- Higher profits

FlexNoC 4 marching ahead

Arteries IP is the world's leading supplier of silicon-proven commercial NoC. On October 31, 2018, it announced its new Arteries IP, FlexNoC version 4. FlexNoC 4 and AI package (FlexNoC 4AI) implement many new technologies that ease the development of today's most complex AI chips, deep neural network (DNNs) and autonomous driving SoCs.

Such massively parallel processors should meet the requirements of high bandwidth for on-chip and off-chip communications. FlexNoC 4 with AI provides the features required for AI chips in an easy-to-use and highly-configurable form.

One user for FlexNoC 4 of Arteries IP is Mobileye. Arteries IP's new processor helps Mobileye create multiple on-chip cache coherent sub-systems and develop heterogeneous programmable hardware accelerators. These sub-systems work together to accelerate speed, increase computational bandwidth, and reduce power consumption of AI and ML algorithms to give near real-time latencies as a car moves.

Applications of FlexNoC 4 are:

- Cloud computing
- Digital home
- The IoT
- Mobiles, cameras, SSD controllers, etc
- Networking
- Storage
- Sensor fusion
- AI for automobiles
- Servers for customers such as Samsung, Huawei, HiSilicon, Mobileye and Texas Instruments

New capabilities in FlexNoC 4 include:

- Automated topology generation for ring, mesh and torus networks

• Multi-cast chip allows for more efficient updates of DNN weights, image maps and other multi-cast data

• Source synchronous communications; the chip helps avoid clock tree synthesis, physical placement and timing closure problems when spanning long distances on AI chips, which can be larger than 400sq mm

• VC-Link; virtual channels allow sharing of long physical links in congested areas of the die while maintaining QoS

• HBM2 and multi-channel memory support

• Up to 2048-bit-wide data support

• Provides long-chip paths

• Provides access to off-chip memory

• Provides homogeneous accelerators

FlexNoC 4 architecture typically models small-world networks (SWNs) and scale-free networks (SFNs) to limit the number, length, area and power consumption of interconnected wires and point-to-point connections. Whether you are using AMBA AXI3, AXI4, AHB, APB, OCB, PIF or a proprietary protocol, Arteries FlexNoC 4 IP reduces the number of wires by nearly half, resulting in fewer gates and a more compact chip floor plan.

Looking ahead

Experts feel that technological advancements are making it necessary to have more sophisticated and an increasing number of CPU cores in a smart processor—and the cores will continue to multiply. FlexNoC 4 processor will push the performance envelope of several sectors and allow several applications and sensor fusion to deliver new experiences, all while extending and saving battery life.

FlexNoC 4 processor is destined to become a reality soon. Rapid increase in consumer demand for multimedia, smart home, growth of Big Data, AI, automobiles, and robotics and more will increase computational speeds. Besides, low power consumption, size and weight are key areas for improvements to be seen in the NoC segment. **EFY**

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